Simulating Collective Transport of Virtual Ants

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What is this paper trying to solve?

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This paper is trying to solve the challenge of creating natural and immersive collective transport simulations that’s realistically adaptable to group size and object weight, as well as external input and interference.

Who in the "real world" cares about this problem?

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- Animators

  who need realistic ant behavior without hand animating every individual

- Roboticists

  who need groups of robots to cooperate without a central controller as seen here

- Stigmergic Systems

  Anyone who wishes to apply stigmergic behavior to a collective of individuals in the technological field

Ants vs Obstacles

How does this paper contribute towards fixing the problem?

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By proposing a stochastic probability model

SEARCH: ants are spawned in at random positions, then individually search for the object until they are within a 2 cm range

APPROACH: ants will directly approach the object until a collision with the object is detected, then will latch itself to the object

TRANSPORT: ants will apply force to move the object. Also divided into 3 states

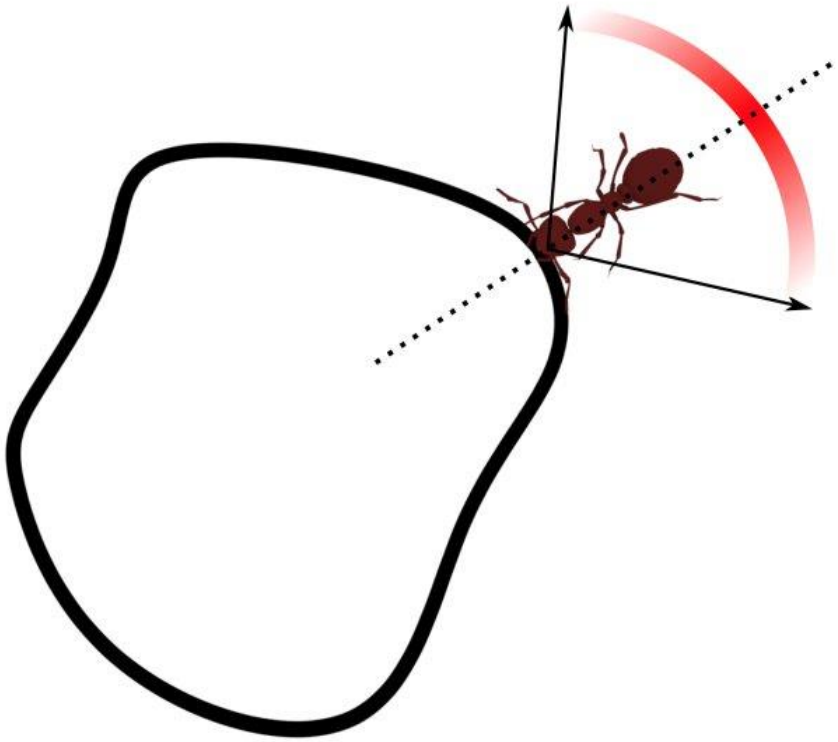
standard: proceeds with the transport function uninterrupted

realign:

reposition:

Realign

refers to the change in the individuals body orientation without letting go of the object, occurs before/more often than reposition and is caused by object weight, friction, and obstacles



Realign: equation

$$P_{realign} = \frac{1}{1 + exp(0.5 - \underbrace{\frac{||\vec{\nu}_o||}{||\vec{\nu}_a||_{max}}}_{\text{maximum ant moving speed}})}$$

^
exponential function

$\vec{\nu}_o$ --> velocity

Realign: target angle

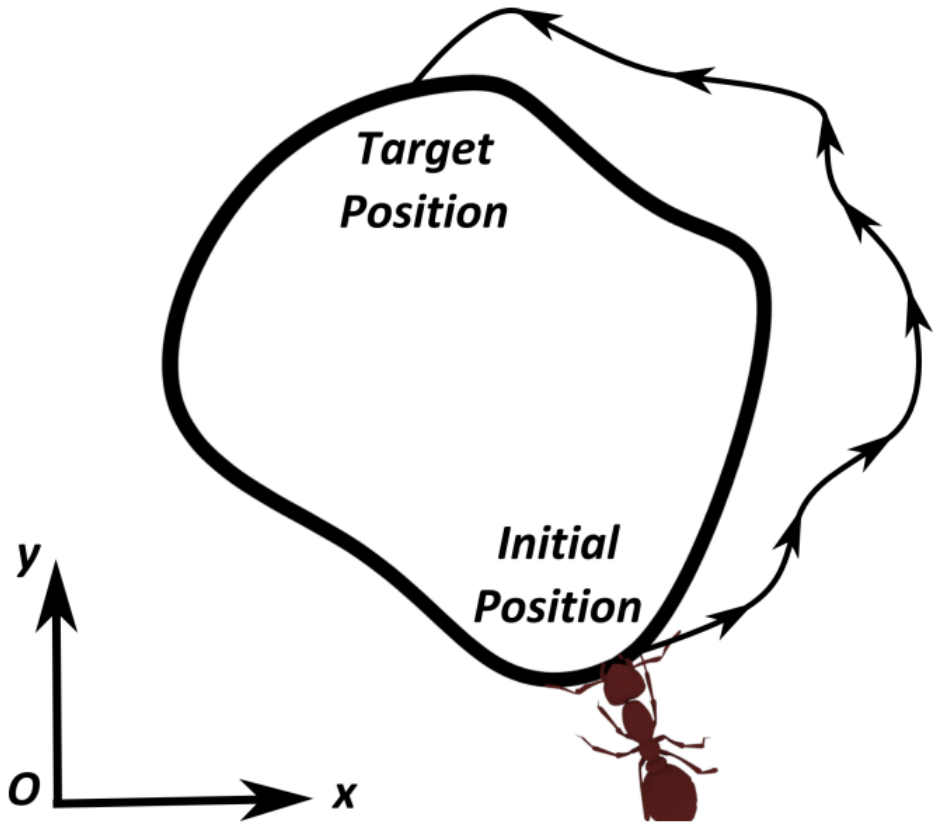
to avoid collisions with the object

$$\theta = N(\theta_{back}, \sigma_{body})$$

the orientation when pulling backwards

Reposition

refers to the individual releasing the object to find another attachment point to grab onto after realignment is unsuccessful



Reposition: equation

$$P_{reposition} = \frac{1}{1 + exp(\frac{t}{t_{max}} - \gamma)} \times \frac{1}{1 + exp(\frac{||\vec{\nu}_a||}{||\vec{\nu}_a||_{max}} - 0.5)}$$

$\vec{\nu}_a$ --> movement velocity of the ant

t, t_{max} --> elapsed time and maximum time from search state to current attachment position

Reposition: movement trajectory

a uniform random distribution between $[D_{min}, D_{max}]$

$$T(t) = C_o(t) + D(t, D_{min}, D_{max}) + C_c(t)$$

outline of the objects shape coordinates

sub level trajectory to prevent collisions with other ants (circular curve with a constant radius)

Stochastic threshold

ensures that probability distribution between 0 and 1 is greater than 98%. If the probability is over the threshold, the individual realigns/repositions itself

$$P_{realign}$$

$$\lambda_a \sim N(\mu_a, \sigma_a)$$

$$P_{reposition}$$

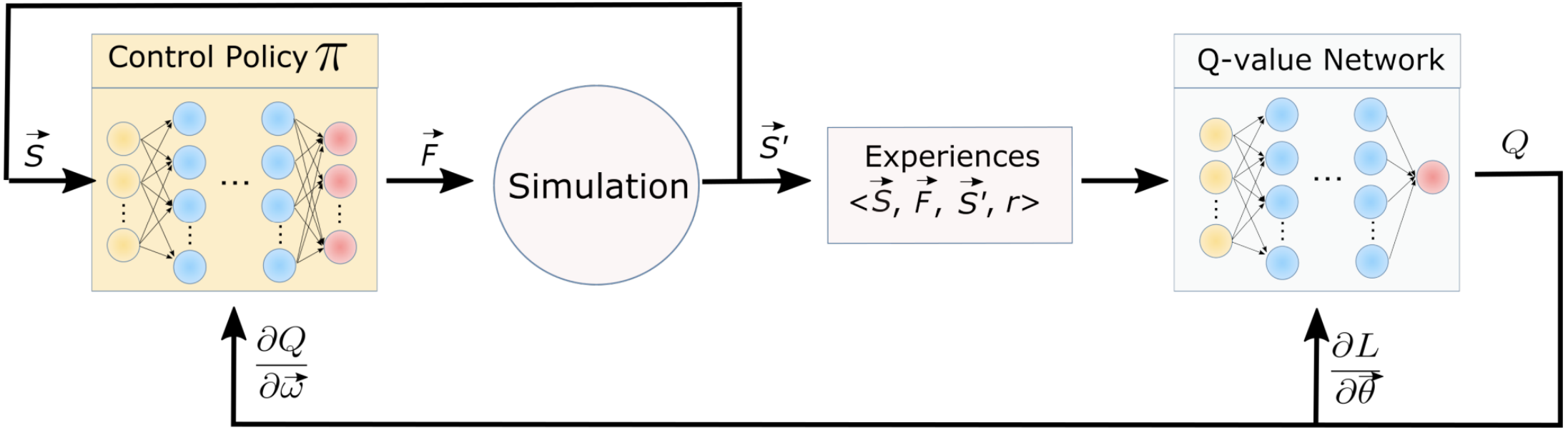
$$\lambda_p \sim N(\mu_p, \sigma_p)$$

$$\mu_a = 0.5 = \mu_p$$

$$\sigma_a = 0.2 = \sigma_p$$

Neural Network Force Policy

The movement force applied when an ant is connected and in transport state.
2 layered



Feed-Forward Neural Network

defines a policy (π) that determines the force (F) to change the current state (S)

Parameter of the decision network

v

$$\pi(\vec{S}, \vec{\omega}) : \vec{S} \rightarrow \vec{F}$$

Velocity of the individual

v

Desired object transport velocity

v

$$\vec{S} = (\vec{\nu}_a, \vec{\nu}_o, \vec{\nu}_o^*)$$

^

Velocity of the object

Q-Value Network

reinforcement learning algorithm to optimize the Feed-forward network (π)

Parameter for the second neural network

v

$$Q(\vec{S}, \vec{F}, \vec{\theta}_Q)$$

Represents the max discounted future reward when F is chosen in state S

Data set for later training--> processes $<\vec{S}, \vec{F}, r, \vec{S}'>$

Q-Value Network: Total future reward

Q-value for a specific time

$$Q_t(\vec{S}, \vec{F}) = r_t + \gamma r_{t+1} + \cdots + \gamma^{n-1}\gamma_{t+n}$$
$$= r_t + \gamma Q_{t+1}(\vec{S}', \vec{F}')$$

γ --> discount factor of the future reward

- If γ =0: policy considers instant reward and disregards the future one
- If γ =1: policy considers future reward
- 0.9 is the preferred balance

r_t --> reward time at *t*

Reward equation for a specific time

$$r = \underbrace{e^{-c_\nu(\vec{\nu}_o^* - \vec{\nu}_o)^2}}_{\text{Minimizes difference between actual and desired velocity}} + \overbrace{e^{-c_\theta(\theta_{back} - \theta_F)^2}}^{\text{Prioritizes backward pulling direction}}$$

Force policy and Q-value network

similar in structure. They're made up of 5 separate layers,
First and last layer: linear-weight neuron
Hidden layer: rectified linear unit

Layers	Force Policy	Q-value Network
Input	9	12
Hidden #1	16	16
Hidden #2	32	32
Hidden #3	16	16
Output	3	1

How are the results evaluated?

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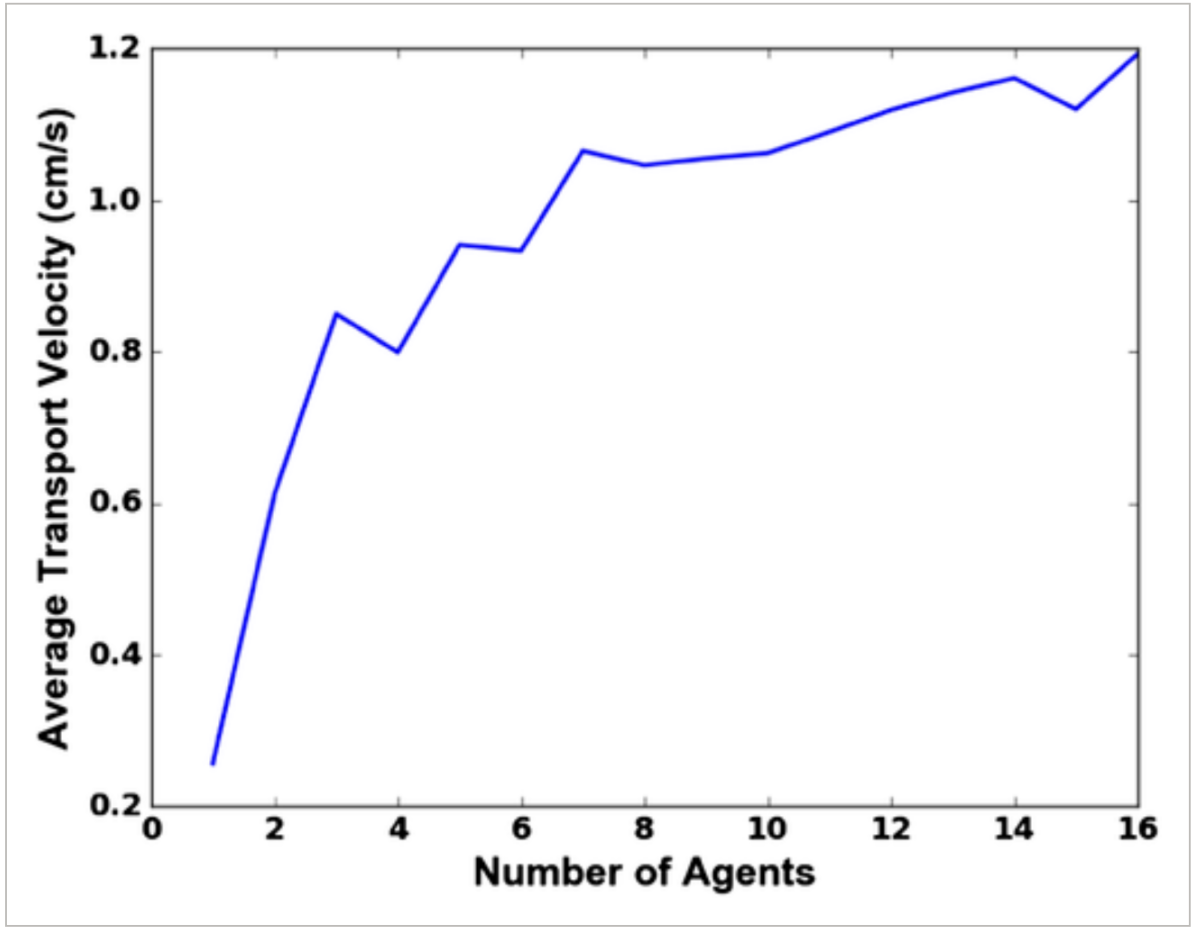
- Results evaluated after 150,000 training iterations (~30 hours) on 8 core system. Results are best showcased in the demo video

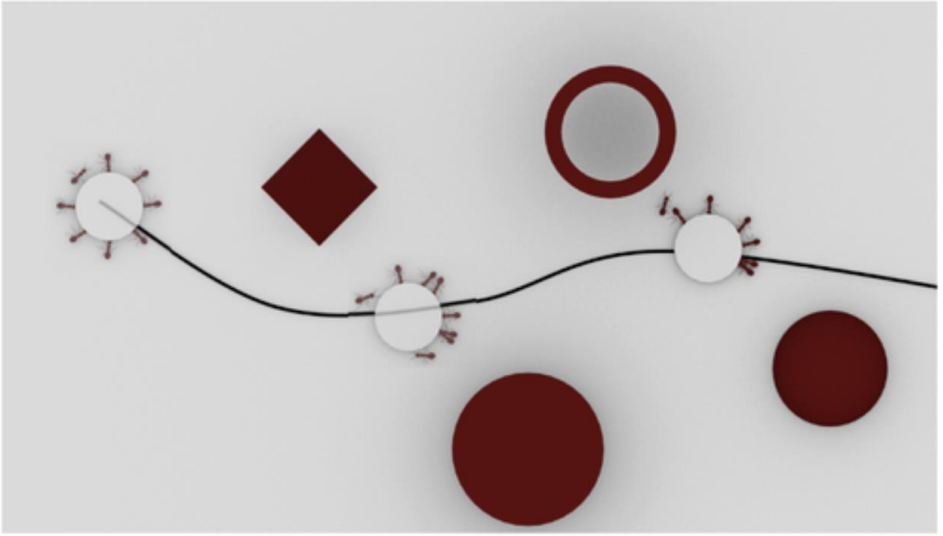
How are the results evaluated? Scalability:

Using the data from the simulations, this graph shows the average transport speed in respect to a number of ants.

- The greater the amount of ants are, the slower the transport sped becomes. This is because most of that extra time is spent trying to locate an appropriate attachment position



How are the results evaluated?
Following a curve:



User defined curves imitate the pheromone paths left by other ants in real life.

- Ants automatically avoid obstacles by following the animator defined path, meaning no extra code needed

How are the results evaluated? Adapting to different objects:

Any object shape works. The outlines are stored as connected line segments which checks for collisions

- Results best shown in the demo video

How are the results evaluated?
External intervention:

In cases of external intervention, when the object is suddenly moved, all ants release it and re-enter Search mode on their own

- Once they find the new location they reattach and resume transport. This recovery is not programmed separately, it comes from the state machine naturally

What are the strengths?

Strengths

- Fully decentralized to imitate stigmergy
- Is scalable from 2-60 individuals
- Works with any object size and defined path
- Gives animators control over the simulation
- Handles interference naturally

What are the weaknesses?

Weaknesses

- Never cross verified with real ants
- Force policy was designed using trial and error
- Frame rate slows down for larger groups
- Does not account for different ant species
- Training time is a very lengthy process

What future work can be done that builds on this paper?

Future Work

- Validation:
  - The simulation can be refined by recording real ant transport footage and using computer vision to extract position and timing data, then comparing it against the simulation to fix any inaccuracies
- Neural network design:
  - The current architecture was built using trial and error with no scientific backing. A more structured approach could produce a better and more generalized force policy
- (Robotics) Full body motion:
  - A future study could build on the stochastic model to account for the full body motion of robots, something the current paper does not address

Conclusion

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The Stochastic model provided in this paper is effective in solving the issue of making the collective transport of ants animation look and behave naturally. The stigmergic behaviors of the ants were successfully replicated using both the model, and the neural network force policy. This method makes it so that the simulation is scalable and adaptable to any object, environment, or group size

Thank you for listening!